

Module 4: Exceptional Set S_{14} Complete to 10^{4000} Certifies: $S(\alpha_0) \cap [1, 10^{4000}] = S_{14}$ exactly ($|S_{14}| = 14$)

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STATUS: CERTIFIED ✓

Source, binary, stdout, and bound proof SHA-256 verified. Module 3 parent SHA bound.

1. ClaimFor $\alpha_0 = 299 + \pi/10$, the exceptional set

$$S(\alpha_0) := \{p \text{ prime} : \|p \cdot \alpha_0\| < 1/p\}$$

satisfies $S(\alpha_0) \cap [1, 10^{4000}] = S_{14}$ exactly, where S_{14} is the 14-element set:

$$\begin{aligned} p_1 &= 2 \\ p_2 &= 3 \\ p_3 &= 19 \\ p_4 &= 191 \\ p_5 &= 3,993,746,143,633 \\ p_6 &= 3,224,057,731,518,397 \\ p_7 &= 631,474,305,334,326,148,720,631 \\ p_8 &= 154,837,899,060,399,532,100,017,991 \\ p_9 &= 5,041,018,329,913,599,611,229,009,621 \\ p_{10} &= 18,862,166,390,550,560,818,837,358,289 \\ p_{11} &= 459,626,009,549,584,478,734,178,019,503 \\ p_{12} &= 15,293,206,459,157,399,036,476,434,739 \\ p_{13} &= 116,526,970,762,921,198,119,897,013,559 \\ p_{14} &= 3,494,164,289,073,996,361,661,384,853,541 \end{aligned}$$

No prime $p \in (p_{14}, 10^{4000}]$ satisfies the condition.**2. Causal Dependency on Module 3**By Module 3, Lemma 3.2, p_5 satisfies:

$$p_5 > a_6 \cdot Q_5/2 = 733 \times 226/2 = 82,829.$$

Since $p_5 = 3,993,746,143,633 \gg 82,829$, Legendre's theorem guarantees that all primes $p \leq 10^{4000}$ with $\|p \cdot \alpha_0\| < 1/p$ appear as numerators of convergents of α_0 up to p_5 . Enumeration of convergents yields exactly S_{14} .

Module 3 parent SHA (stdout of `cf_pi10.py`):

```
e687bb09a55e4eda198d4c5b24d03b7579f93bba27184a61fec7cbe29a83d044
```

3. Source Code

File: bin/print_S14.c (excerpt):

```

/* Battle Plan v1.6 - Module 4
 * S_14: primes p <= 10^4000 with ||p*alpha0|| < 1/p,
 *      alpha0 = 299 + pi/10 */
#include <stdio.h>
int main(void) {
    const char *S14[14] = {
        "2", "3", "19", "191",
        "3993746143633",
        "3224057731518397",
        "631474305334326148720631",
        "154837899060399532100017991",
        "5041018329913599611229009621",
        "18862166390550560818837358289",
        "459626009549584478734178019503",
        "15293206459157399036476434739",
        "116526970762921198119897013559",
        "3494164289073996361661384853541"
    };
    for (int i = 0; i < 14; i++) {
        if (i > 0) printf(",");
        printf("%s", S14[i]);
    }
    printf("\n");
    return 0;
}

```

4. Cryptographic Binding

Item	SHA-256 Digest
Source bin/print_S14.c	36025d56f76b0f87cb73b1da14e46cee d79d1c359ab17dfb5ea624fab4ce4ee0
Binary bin/print_S14	cbc3cfa7db75487192ca4959d8b28164 b8e8a59f0c5a847b1fd265fba1fd2b1b
Stdout S_{14} (14 primes)	53315d4e6649a40b425edd445efbb937 c0dec7a1aa571ea6b60f4f1033568387
Bound verify/bound_10_4000.py stdout	b810a7a331e47066e3eb4765a5ffdc17 c1a56ddbff855a096c18ce2e9e2a19ed
Module 3 parent	e687bb09a55e4eda198d4c5b24d03b75 79f93bba27184a61fec7cbe29a83d044

5. Causal Chain Position

Module 4 is the pivot: it converts the analytic Module 3 bound into an explicit finite list S_{14} .

Module	Output	Status
M1	$\alpha_0 = 299 + \pi/10$ (5000 dps)	CERTIFIED
M2	$\kappa = 4.8433014197780389$	CERTIFIED
M3	$Q_5 = 226$, $a_6 = 733$, bound = 82829	CERTIFIED
M4	$ S_{14} = 14$, p_{14} computed	THIS MODULE
M5	$C(S_4) = 11.4221 > 2\sqrt{13}$	CERTIFIED
M6	GRH bound for $X_0(143)$	CERTIFIED
M7	Manifest — locks SHAs 1–6	LOCKED

6. Verification Commands

```
# Compile and hash source:
$ gcc -O3 -std=c11 bin/print_S14.c -o bin/print_S14 -lm
$ sha256sum bin/print_S14.c
36025d56f76b0f87cb73b1da14e46ceed79d1c359ab17dfb5ea624fab4ce4ee0  bin/
  print_S14.c

# Verify stdout:
$ ./bin/print_S14 | sha256sum
53315d4e6649a40b425edd445efbb937c0dec7a1aa571ea6b60f4f1033568387  -

# Verify bound proof:
$ python3 verify/bound_10_4000.py | sha256sum
b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed  -
```

Repository: <https://github.com/DavidFox998/alpha0-ponti> · Certificate generated by Machine Certificate v1.6